



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

Digital Forensics

System Forensics, Investigation, and Response

Lesson 6 Recovering Data

Learning Objective

- Explain techniques for hiding and scrambling information as well as how data is recovered.

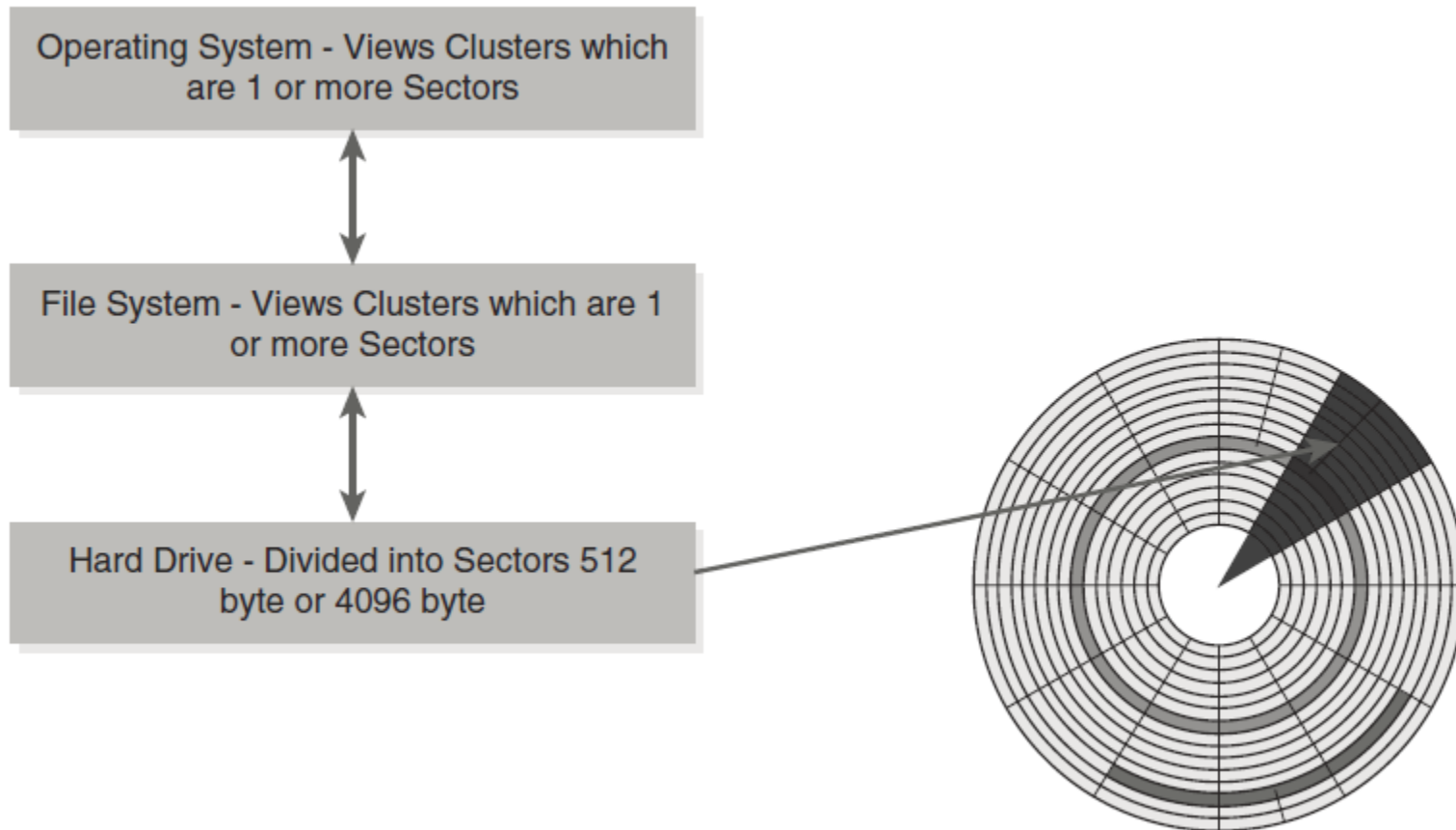
Key Concepts

- Undeleting data
- Recovering information from damaged drives

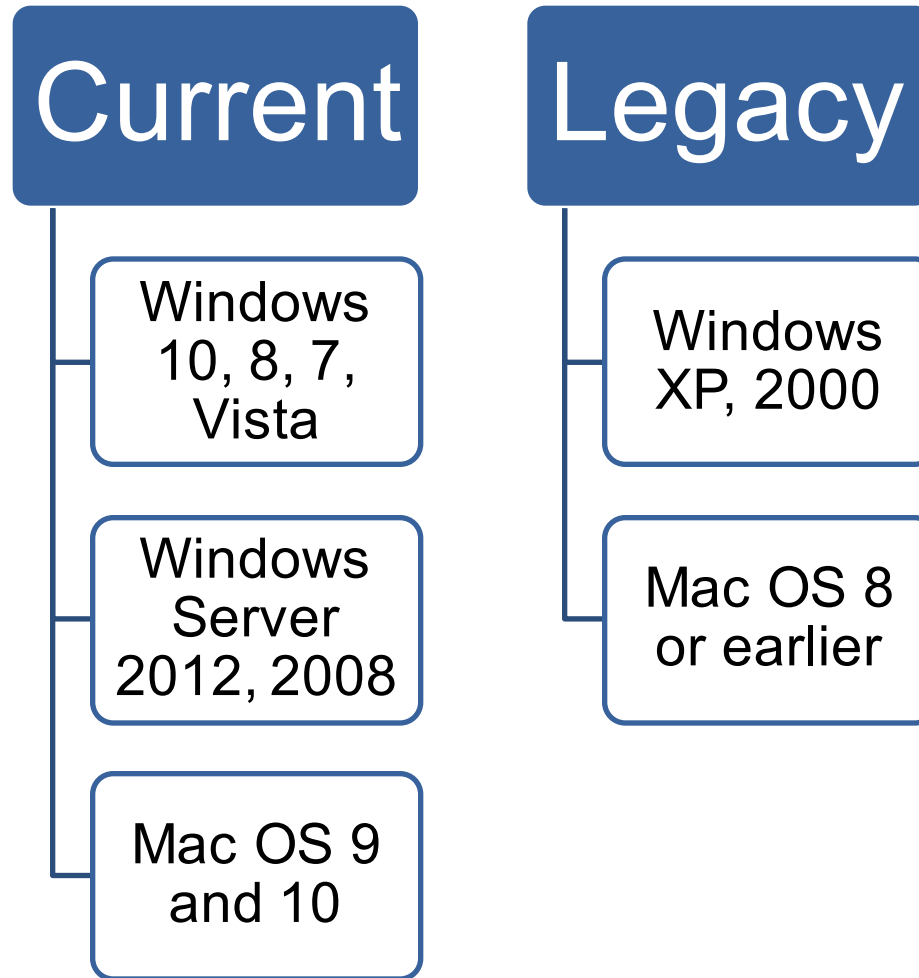
Undeleting Data

- Criminals who are not very technically savvy think that deleting a file will keep authorities from discovering it
- Expect that evidence will frequently be deleted from computers you examine

File Systems and Hard Drives



Operating Systems



Windows

- FAT16 and FAT32 used in pre-Windows 2000 versions
- NTFS file system in use since Windows 2000
- Uses a table to map files to specific clusters where they are stored on the disk

Storing a File in Windows (FAT/FAT32)

Record cluster number for next cluster

Add EOC if at end of chain

Mark bad, reserved, open clusters

Deleting a File in Windows (FAT/FAT32)

- When a file is deleted, data not removed from disk
- FAT is updated to reflect clusters no longer in use
- New data saved to those clusters may overwrite old information

NTFS Fundamental Files

MFT

- Master File Table
- Describes all files on the volume

Cluster
bitmap

- A map of all the clusters on the hard drive

Storing a File in Windows (NTFS)

MFT contains one base file record for each file and directory

MFT serves same purpose as FAT

Cluster bitmap file maps all clusters on disk

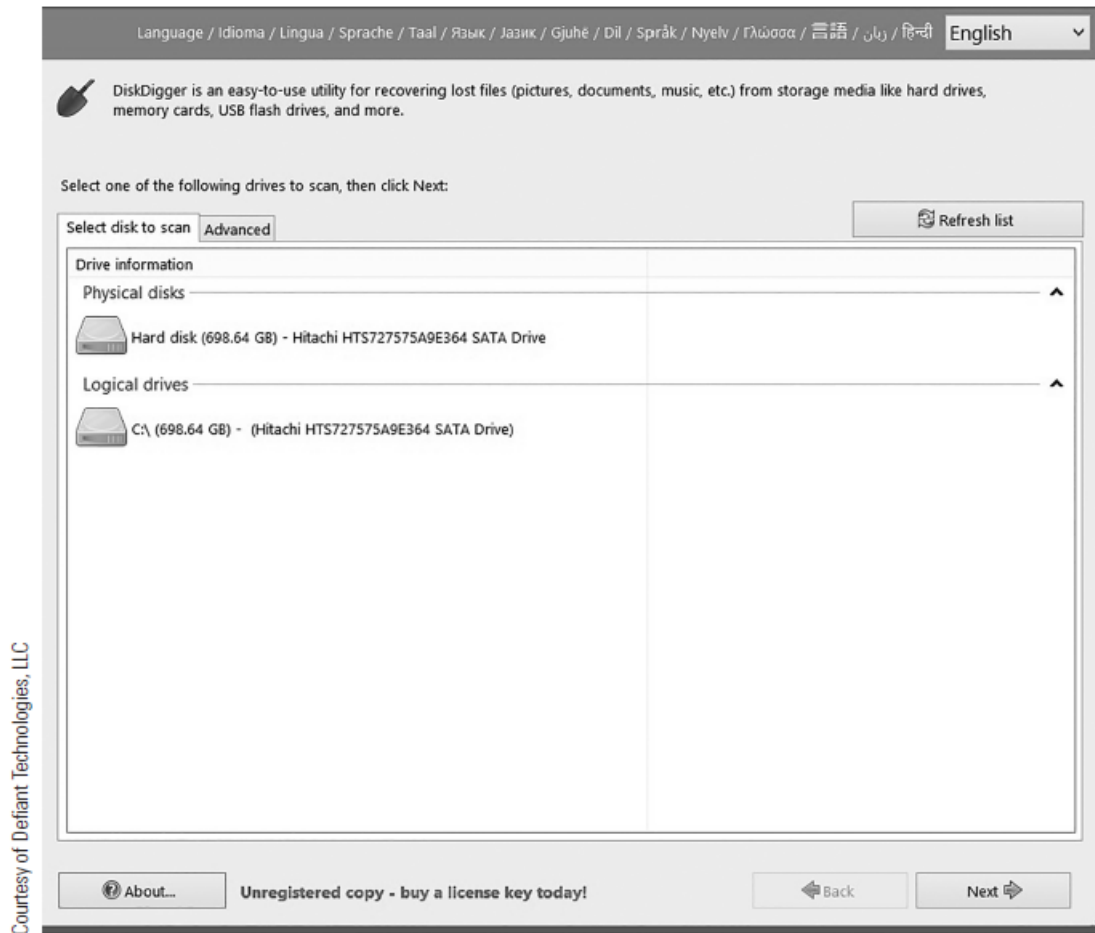
Deleting Files in Windows (NTFS)

- When a file is deleted, data not removed from disk
- Clusters are marked as deleted and “moved” to Recycle Bin
- When Recycle Bin is emptied, clusters marked as fully available
- Filename in the MFT is marked with a special character that means the file has been deleted

DiskDigger

- Free and commercial versions
- Free version recovers files one at a time

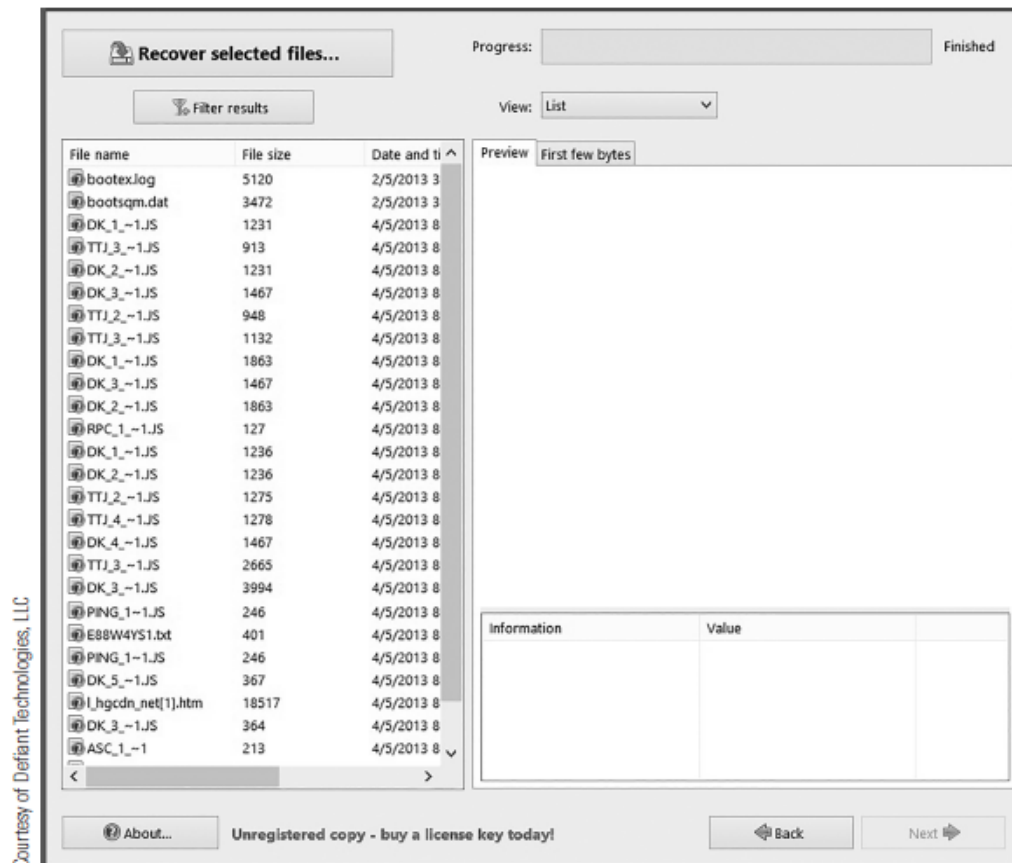
DiskDigger: Main Screen



DiskDigger: Starting Data Recovery



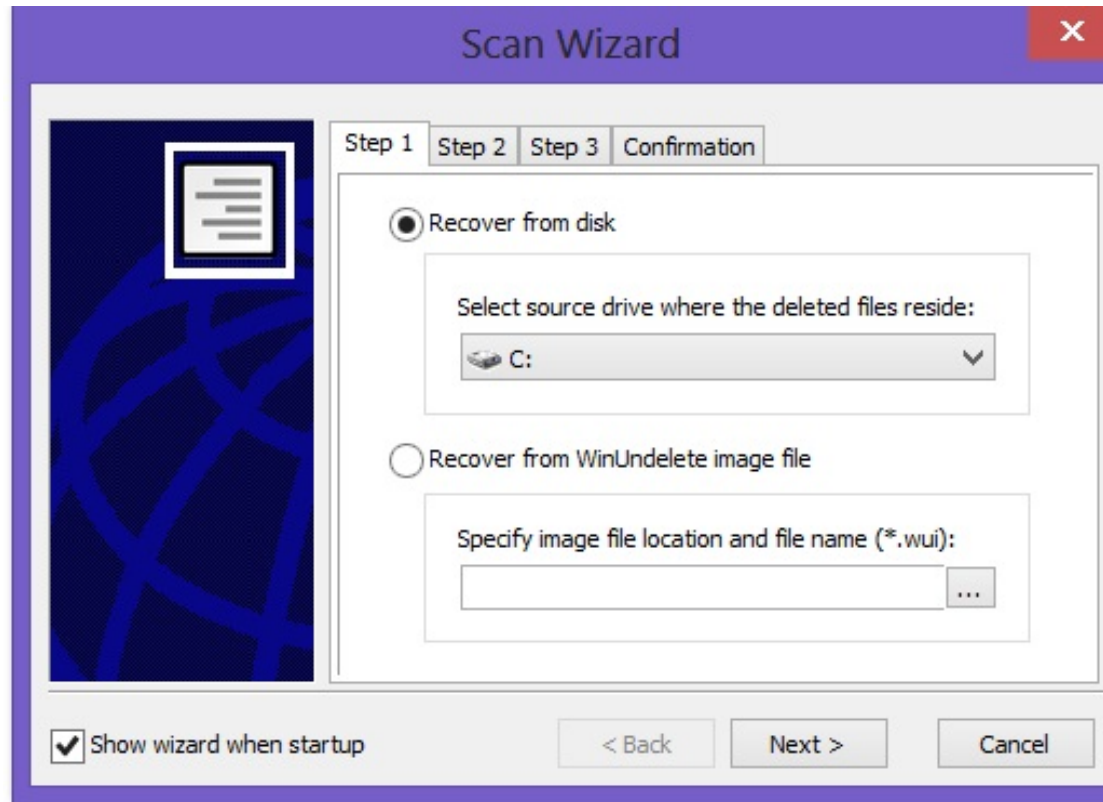
DiskDigger: Recovering an Individual File



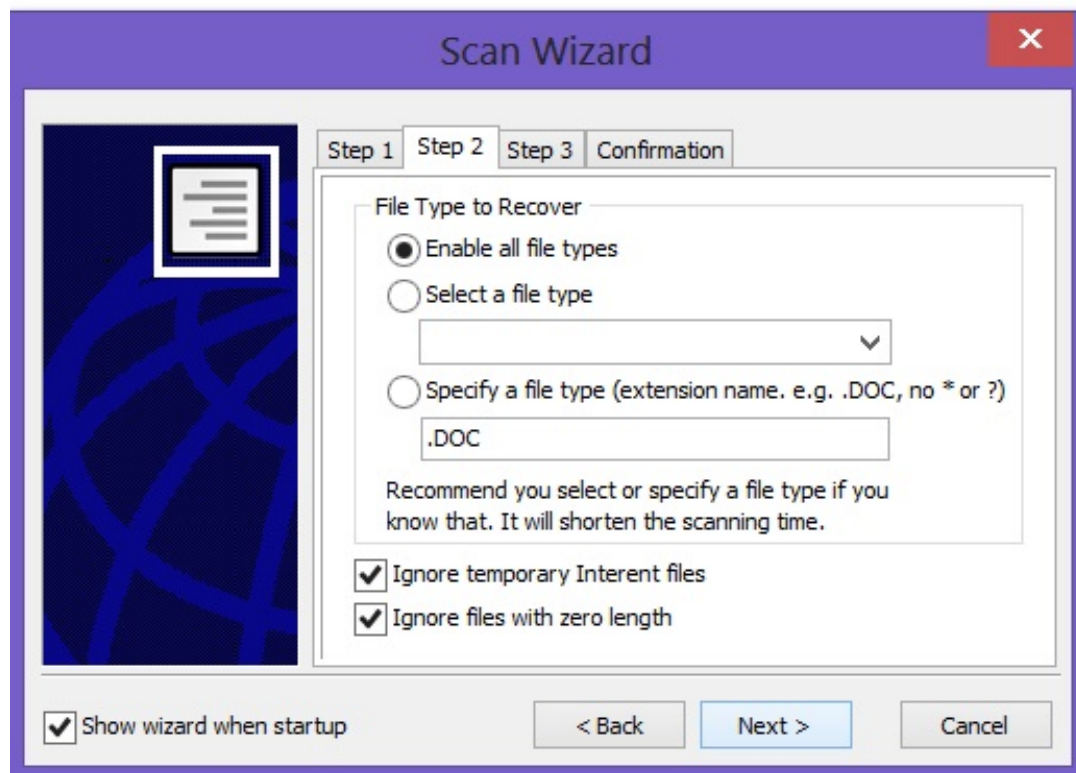
WinUndelete

- Easy to use
- Wizard-driven

WinUndelete

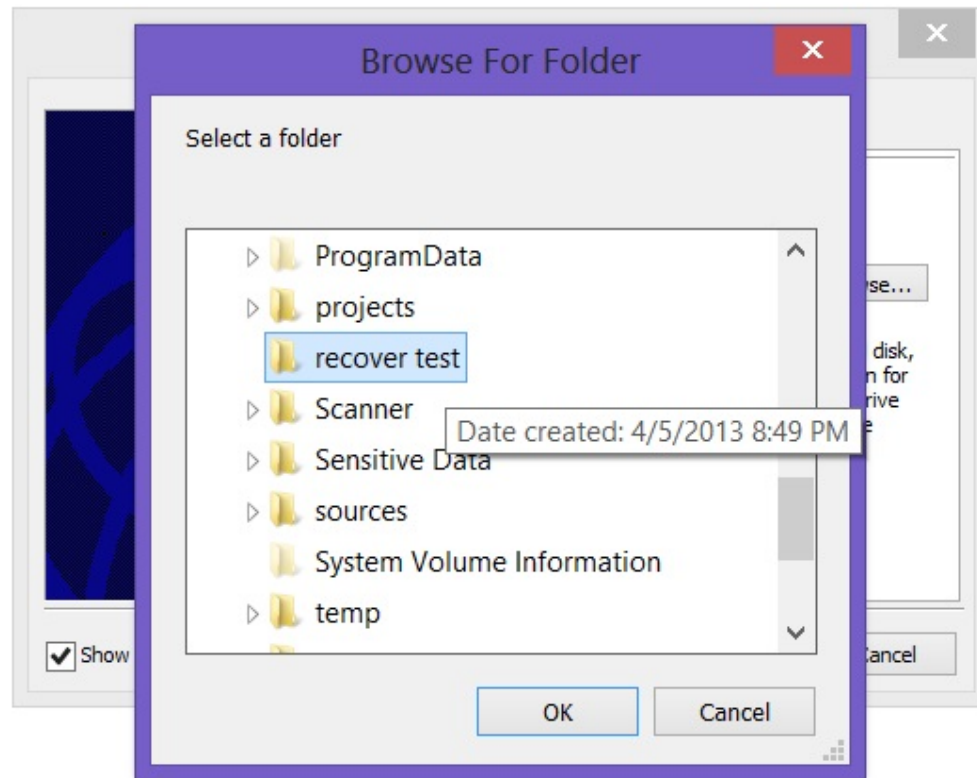


WinUndelete



Courtesy of WinRecovery Software

WinUndelete

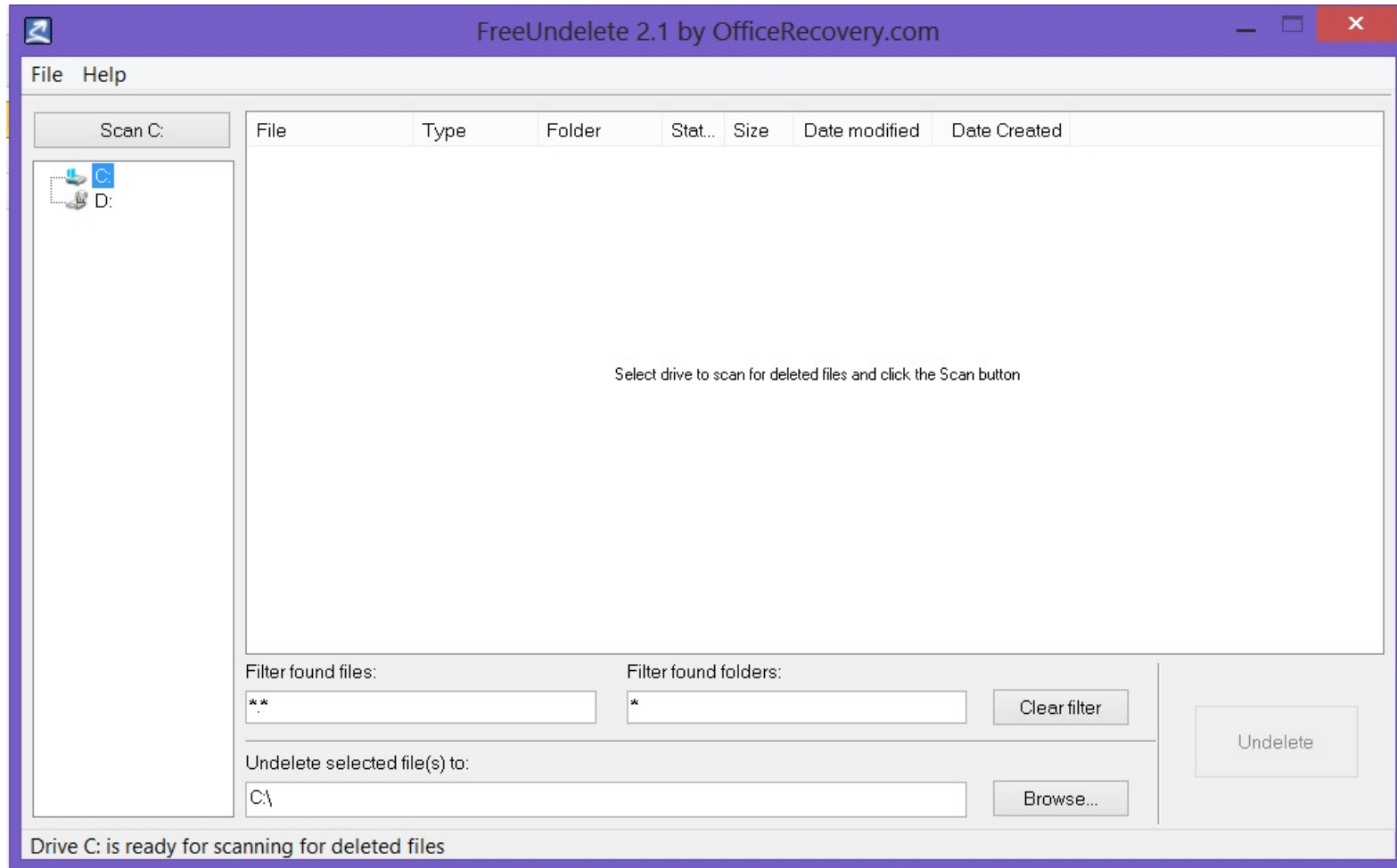


Courtesy of WinRecovery Software

FreeUndelete

- Free tool for personal use
- Commercial version available

FreeUndelete



Courtesy of Recoveronix Ltd.

OSForensics

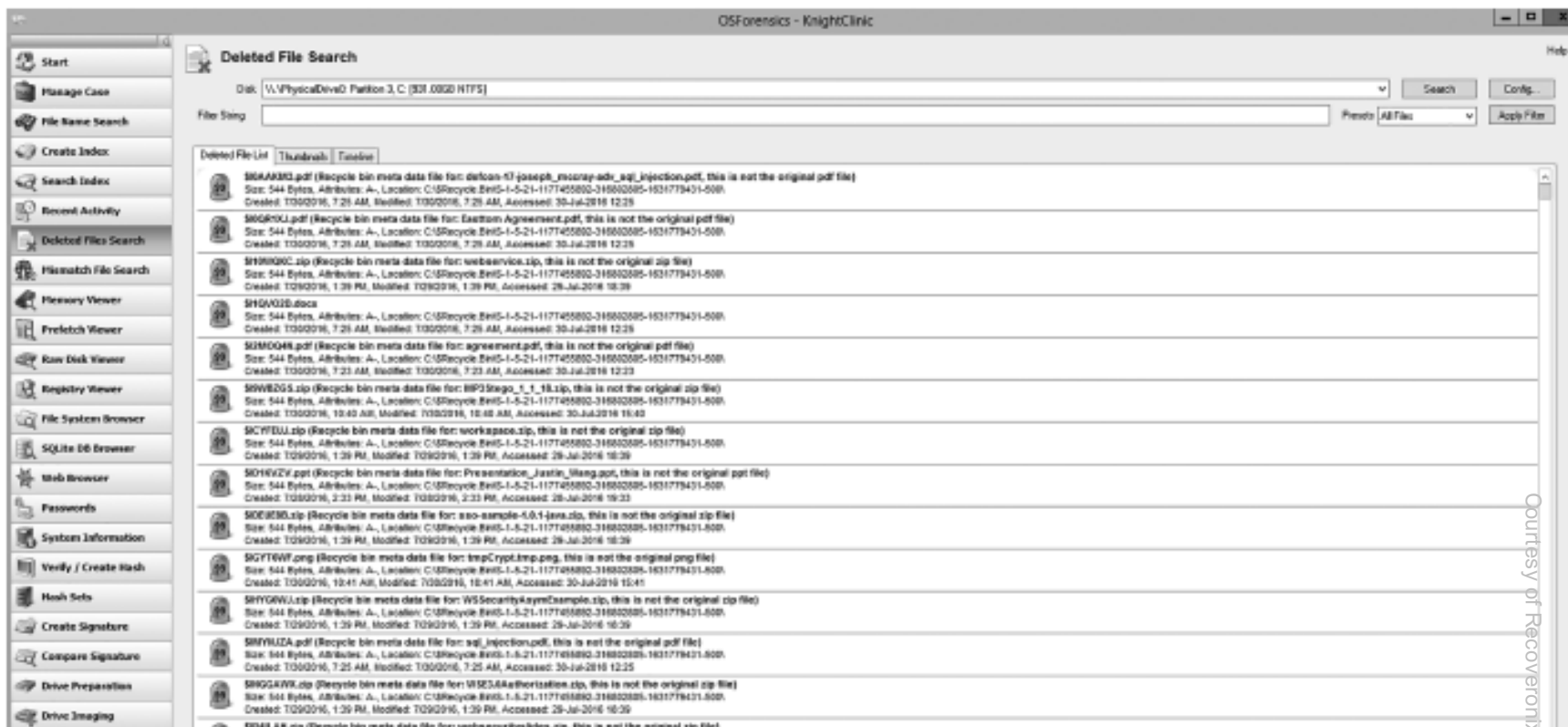
- A robust forensics tool that also provides for undeletion
- Undelete from a mounted image or from a live system

OSForensics Deleted Files Search



Courtesy of Recoveronix Ltd.

OSForensics Deleted Files Results



Courtesy of Recoverit Ltd.

Linux

- File systems
 - ext3
 - ext4

Storing a File in Linux

- Stores files in contiguous blocks
- Blocks sometimes need to be extended
- Exact size of blocks depends on parameters used with the command that creates the partition
- Uses inodes and soft links

Deleting a File in Linux

- Inode hard link is integral
- Inode links directly to a specific file
- OS keeps a count of references to each hard link
- When reference count reaches zero, file is deleted

Recovering a File in Linux

- Move system to single-user mode with *init 1* command
- Use *grep* to search for and recover files
- Example:
 - ```
grep -i -a -B10 -A100
'forensics' /dev/sda2 >
file.txt
```

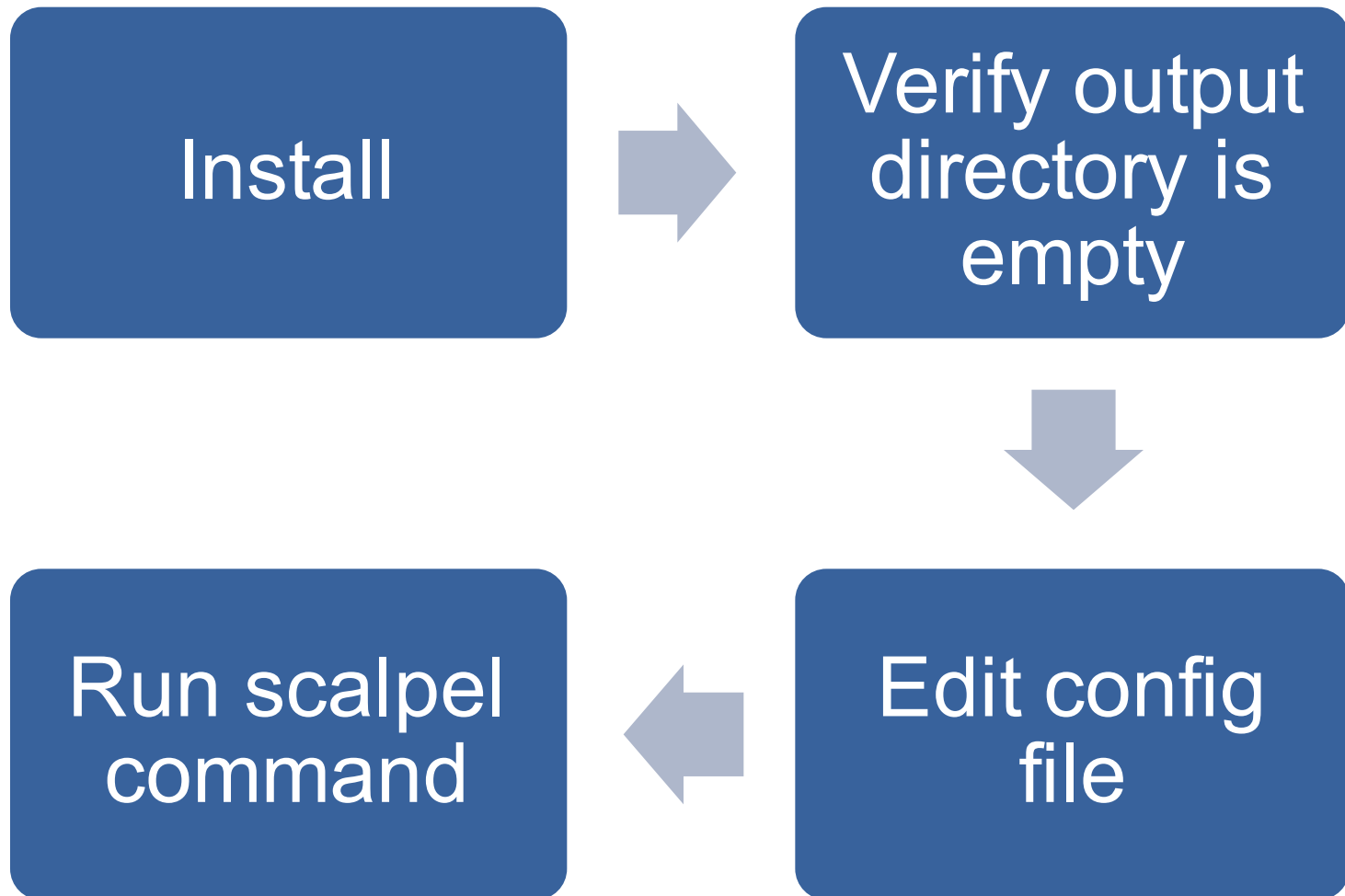
# extundelete

- Works with both ext3 and ext4 partitions in Linux
- Uses shell commands
- Example: To restore all deleted files from sda1 partition:
  - `extundelete /dev/sda4 --restore-all`

# Scalpel

- Works with Linux and Mac OS
- Possible to compile source code to work in Windows

# Scalpel (Cont.)



# Scalpel (Cont.)

1. Install the tool.
2. In the configuration file `/etc/scalpel/scalpel.conf`, uncomment the specific file format you want to recover.
3. Run the following command:

```
sudo scalpel
[device/directory/file name] -o
[output directory]
```

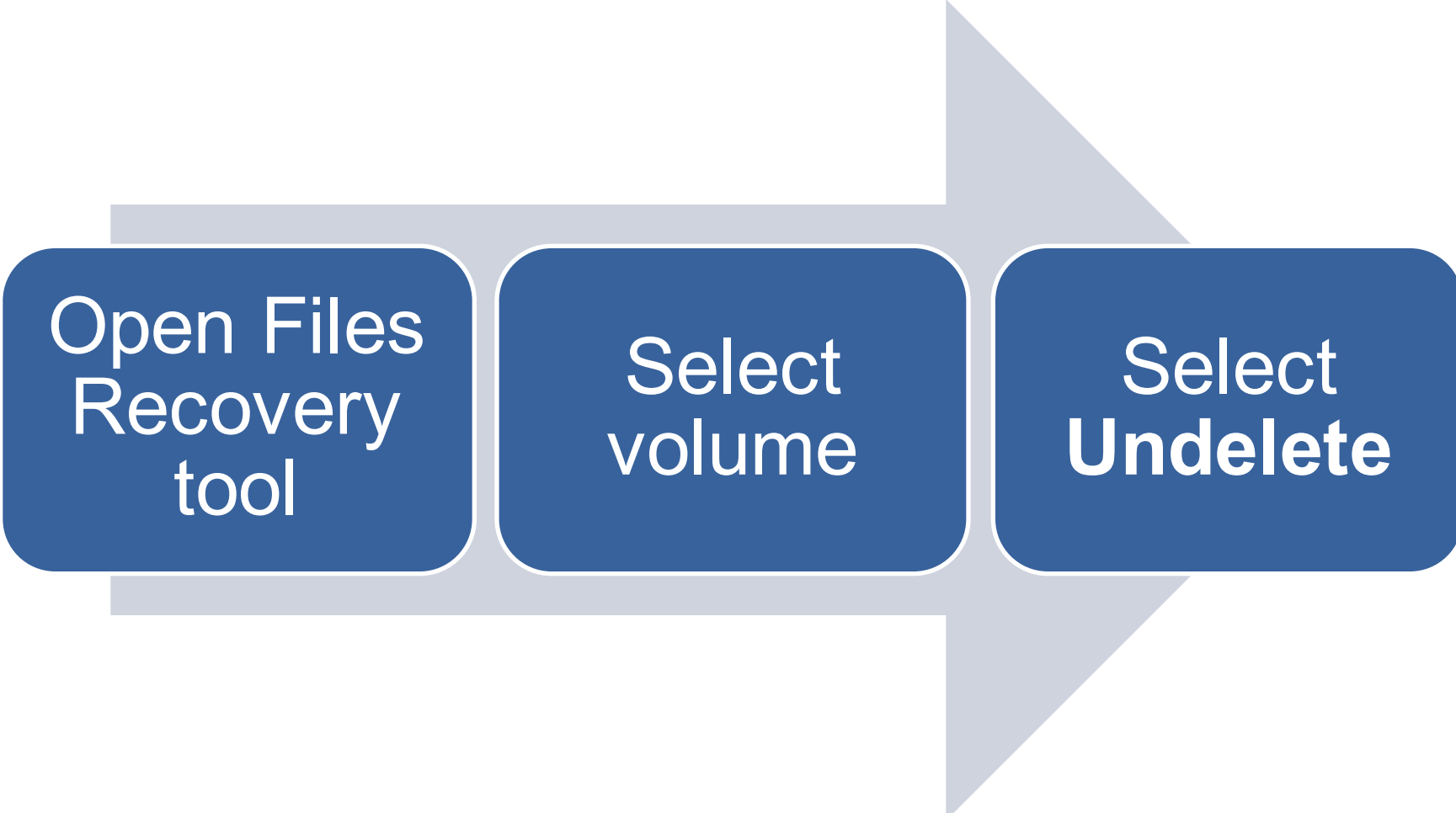
# Macintosh

- Macintosh OS X and later versions are based on FreeBSD
  - A UNIX clone, much like Linux
- Mac OS X uses HFS+, or Hierarchical File System Plus
- Earlier versions of Macintosh used HFS

# MacKeeper

- Recovers deleted files on Macintosh computers
- Free, fully functional trial version available

# MacKeeper

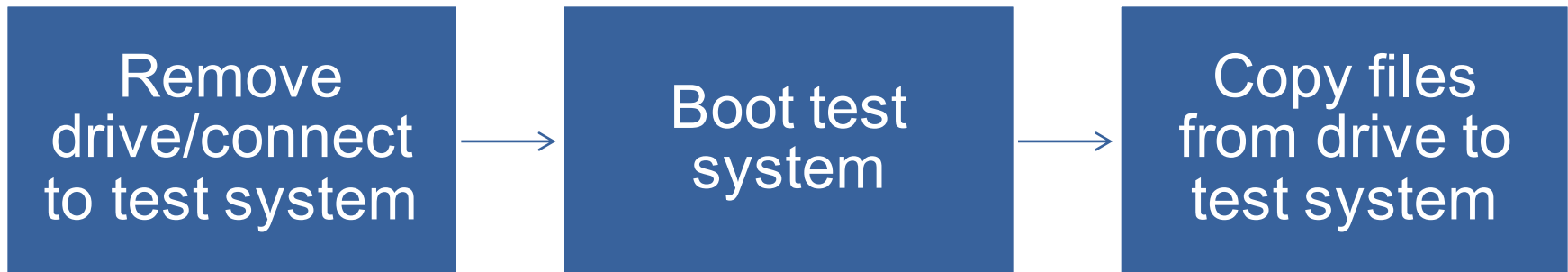


Open Files  
Recovery  
tool

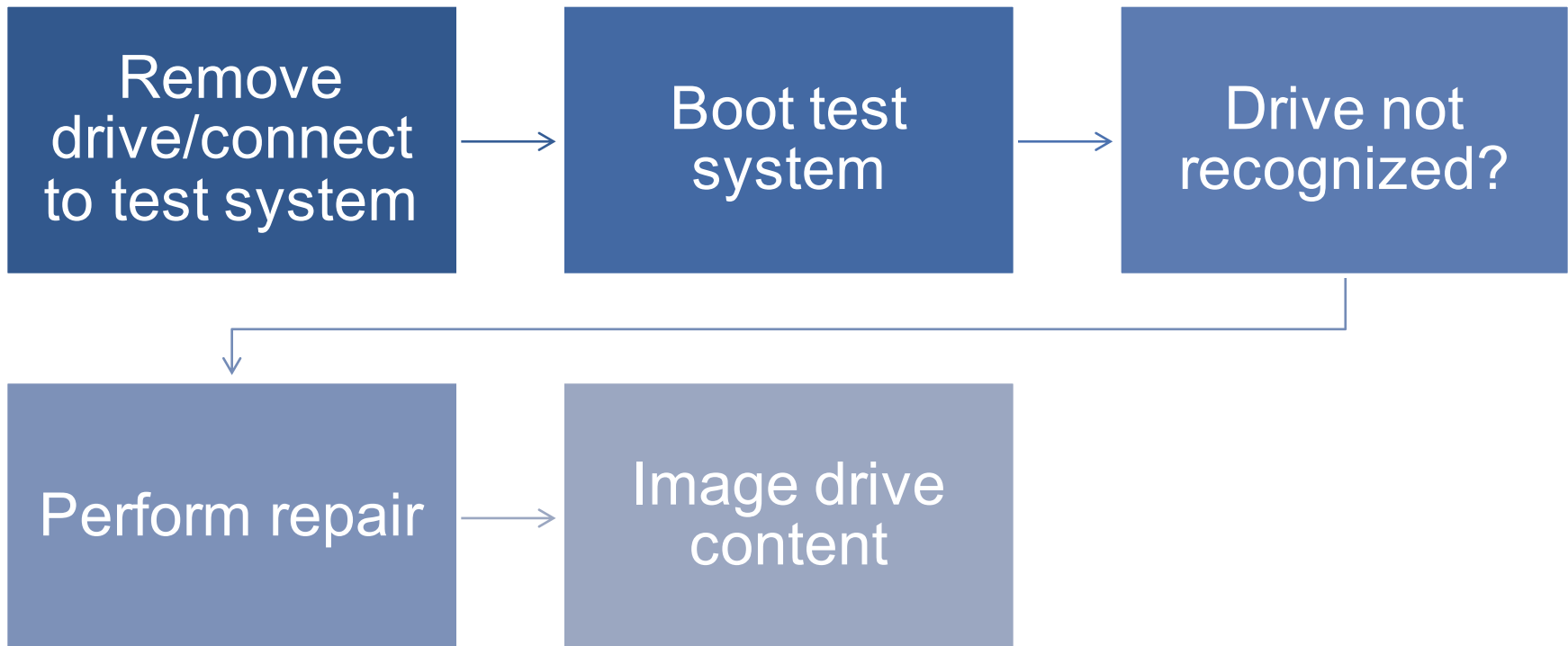
Select  
volume

Select  
**Undelete**

# Recovering Information from Damaged Media



# Recovering Information from Damaged Media (Cont.)



# Attempting Local Repair

Replace printed circuit board

Replace read/write head assembly

Transfer disk platters to healthy drive

# Recovering After Logical Damage

- Logical damage
  - May prevent host operating system from mounting or using the file system
  - May cause system crashes and data loss
  - May be caused by power outages, or turning off a machine while it is booting or shutting down

# Recovering After Logical Damage (Cont.)

- Microsoft Windows: chkdsk
- Linux: fsck
- Mac OS X: Disk Utility
- The Sleuth Kit
- TestDisk

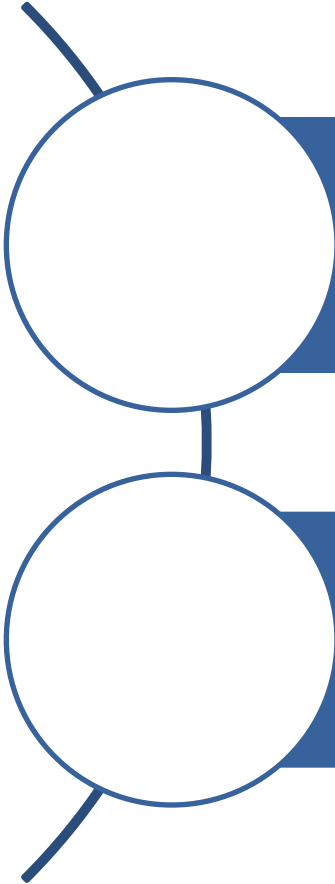
# Preventing Logical Damage

- Journaling file systems
- Use a consistency checker
- Use disk controllers with battery backups

# Consistency Checking

- Involves scanning a disk's logical structure to ensure that it is consistent with its specification
- Verifies that dot (.) and dot-dot (..) entries point to correct directories
- Checkers include chkdsk and fsck

# Consistency Checking Problems

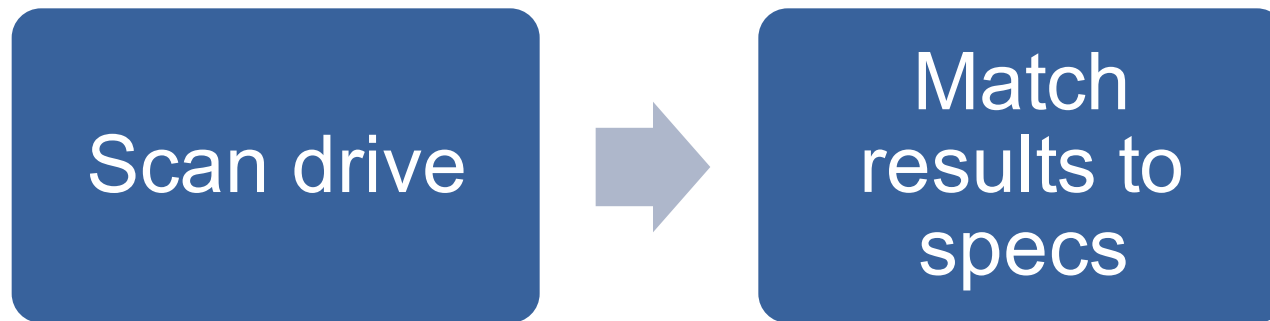


Can fail if file system is highly damaged

Chkdsk utility might delete files that are out of place or unexplainable

# Zero-Knowledge Analysis

- Few assumptions made about state of the file system

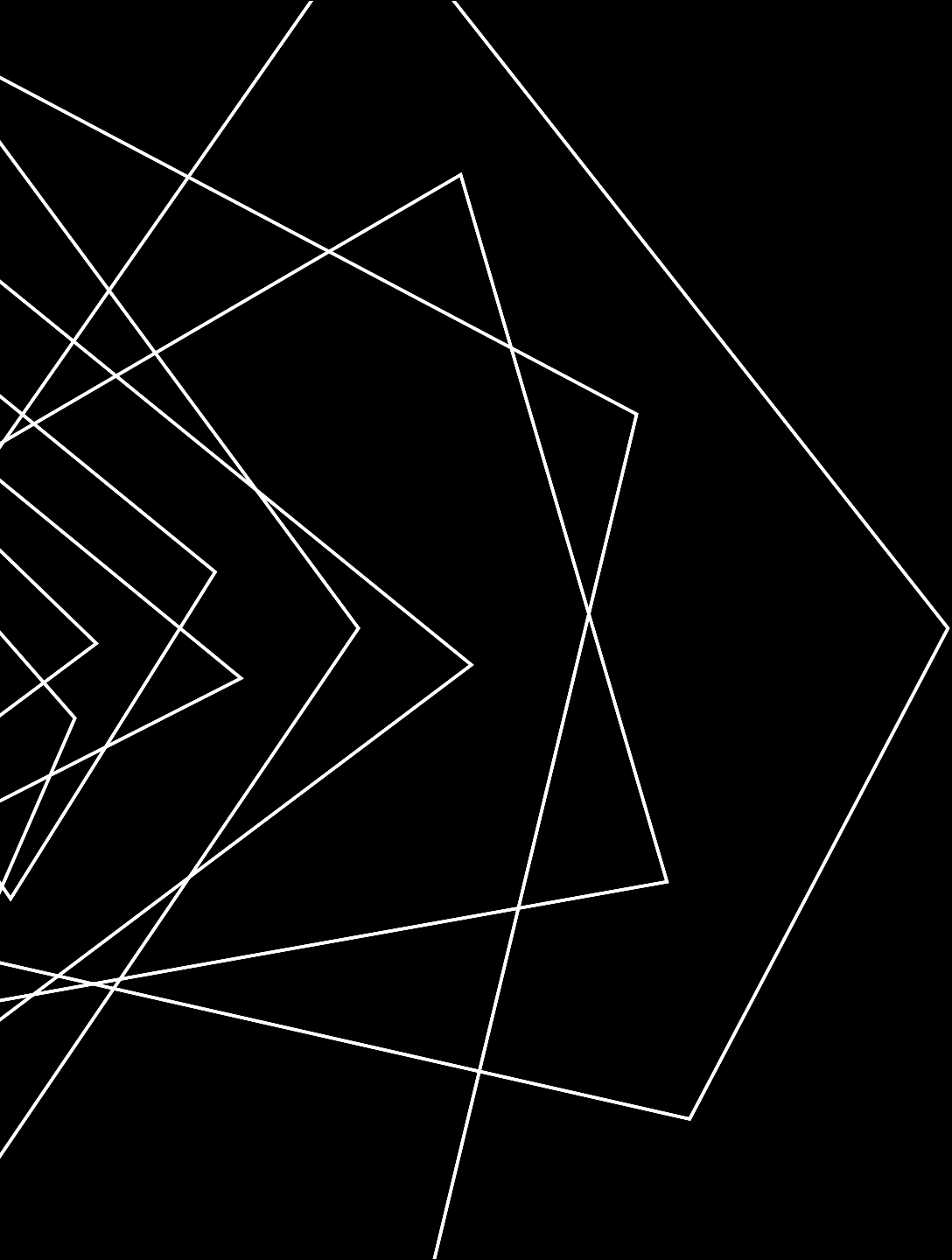


# File Carving

- Can use file carving on a file that's only partially recovered
- Works on any file system
- Is often used to recover data from a disk where there has been some damage or where the file itself is corrupt
- File carving utilities look for file headers and/or footers, and then pull out data s found between the two boundaries
- One popular file carving tool is Scalpel

# Summary

- Undeleting data
- Recovering information from damaged drives



# THANK YOU

Yau Ka Cheung

[Yauka.Cheung@tamuc.edu](mailto:Yauka.Cheung@tamuc.edu)

ACB1 Room 335